

Human-Centered Design Recommendations for LLM-as-a-Judge

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Abstract

Traditional reference-based metrics, such as BLEU and ROUGE, are less effective for assessing outputs from Large Language Models (LLMs) that produce highly creative or superior-quality text, or in situations where reference outputs are unavailable. While human evaluation remains an option, it is costly and difficult to scale. Recent work using LLMs as evaluators (LLM-as-a-judge) is promising, but trust and reliability remain a significant concern. Integrating human input is crucial to ensure criteria used to evaluate are aligned with the human’s intent, and evaluations are robust and consistent. This paper presents a user study of a design exploration called EvaluLLM, that enables users to leverage LLMs as customizable judges, promoting human involvement to balance trust and cost-saving potential with caution. Through interviews with eight domain experts, we identified the need for assistance in developing effective evaluation criteria aligning the LLM-as-a-judge with practitioners’ preferences and expectations. We offer findings and design recommendations to optimize human-assisted LLM-as-judge systems.

1 Introduction

Recent advancements in Large Language Models (LLMs) challenge traditional methods of assessing natural language generation (NLG) quality, as known metrics, such as BLEU (Papineni et al., 2002) and ROUGE (Lin, 2004), fall short for creative tasks. The diverse and expanding ca-

pabilities of LLMs (Liang et al., 2022) present a selection challenge for practitioners, requiring evaluations of extensive outputs across contexts like summarization and retrieval-augmented generation (RAG). The subjective and use case-specific nature of emerging NLG tasks often demands human review, making the evaluation process hard to scale without suitable automatic metrics. While experts can perform evaluations, this is costly and impractical for rapid iteration in early development stages. (Gehrmann et al., 2023).

One potential solution to these challenges is to leverage the capabilities of LLMs to aid in the evaluation process. Despite not always being accurate, LLMs have the potential to significantly reduce the workload by identifying outputs where they are not confident, thus indicating where human input may be required. Additionally, LLMs can assist practitioners in identifying and customizing criteria specific to their use case—such as, for example, faithfulness to contextual information, naturalness of the conversation, and succinctness—with which they wish to conduct their evaluations. This customization enables a more targeted and effective assessment of model outputs, tailored to the specific requirements of their tasks. In this paper, we present results from a user study of EvaluLLM (Desmond et al., 2024), a tool designed to facilitate the evaluation of model outputs. EvaluLLM simplifies how practitioners choose LLMs by offering a quick way to assess and compare their performance across various tasks. This method accelerates the

development of evaluation criteria and helps manage the growing variety and capabilities of LLMs.

To understand the challenges and user needs in model evaluation that leverage LLM-as-a-Judge to automate the process, we conducted formative, semi-structured interviews with 8 practitioners (data scientists, software engineers, and AI engineers) who have been involved in model performance evaluation projects over the past year. Our interviews revealed various challenges and needs. For instance, practitioners highlighted the necessity for rapid performance comparison across different setups, the importance of defining evaluation criteria (e.g., structured and customizable templates aligned with specific use cases), and strategies for effectively integrating LLM-as-a-Judge into their workflow (e.g., starting with a small subset of data before scaling up). In this paper, we present the following contributions:

- We describe EvalLLM (Desmond et al., 2024), an LLM-Assisted evaluation tool that enables users to select multiple models, define custom metrics for NLG evaluation, and review the results while providing feedback to observe the agreement between human and AI evaluations.
- We present qualitative findings from interviews with domain experts (N = 8) revealing challenges and user needs for model evaluation workflows including LLM-as-a-judge.
- We make design recommendations and provide example feature designs to enable users to define criteria interactively, ensuring transparent and rapid access to LLM-as-a-judge’s preferences while balancing trade-offs across multiple dimensions in a self-consistent manner.

2 Related work

LLMs trained to follow instructions can generate results that surpass the quality of data produced by humans. This makes it increasingly challenging to assess the quality of natural language generation (NLG) outputs (Liang et al., 2022) (Xiao et al., 2023) (Liu et al., 2023b). Traditional reference-based metrics, such as ROUGE (Lin, 2004) and BLEU (Papineni et al., 2002), might not effectively capture the essence of LLM outputs, especially in scenarios where the output space is broad and varied. This means multiple different outcomes

can all be valid, making it nearly impossible to create sufficiently comprehensive reference sets. Consequently, these metrics may not be reliable indicators of NLG output quality, as they often demonstrate a low correlation with human judgments (Freitag et al., 2022).

Recent advances highlight LLMs’ potential as customizable judges, (Liu et al., 2023a) (Wang et al., 2023a) (Zheng et al., 2023) capable of adapting to various tasks beyond traditional evaluation methods. Techniques like G-Eval (Liu et al., 2023a) use chain-of-thought prompting and form-filling to assess NLG quality, while GPTScore (Fu et al., 2023) evaluates using conditional token probabilities, enhancing scoring granularity. AlpacaEval (Li et al., 2023) (Yuan et al., 2024) compares model win rates, and Prometheus (Kim et al., 2023a) is a fine-tuned LLM specifically designed for evaluation tasks. These methods align closely with human preferences, especially in creative tasks, emphasizing LLMs’ ability to mimic human judgment. Their effectiveness relies on tailored prompt design and user-defined criteria for precise evaluations. While not part of this paper, in our own work, we have also done comprehensive benchmarking of human agreement of different LLM-as-a-judge approaches for different use cases and we found that depending on use case, LLMs as judges, and judging approach, we were able to achieve good results. Note that this is often a hard problem for humans too and inter-rater reliability can be a good reference.

Previous research has investigated using expert-labeled data to develop custom evaluation metrics like AUTOCALIBRATE (Liu et al., 2023b), but this method is limited by the availability of such data. For reference-free evaluations, interactive human involvement is preferable, allowing users to refine criteria effectively by reviewing outputs. ConstitutionMaker (Petridis et al., 2023) enables feedback on model outputs to iteratively refine prompts, focusing more on AI prototyping than evaluation. Other tools like Zeno (Cabrera et al., 2023), the What-If Tool (Wexler et al., 2019), and Errudite (Wu et al., 2019) help identify model vulnerabilities by analyzing specific data segments. EvalLLM (Kim et al., 2023b) allows users to define criteria interactively, using LLM-as-a-judges for output ratings, although this can be limited by LLM reasoning capabilities (Zheng et al., 2023). Our study builds on these insights, proposing a system where practitioners define criteria in natural language for LLMs to perform pairwise comparisons, enhancing

trust through a "human-in-the-loop" blind review process that eliminates the need for expert data.

3 EvaluLLM

To explore how to support users in developing their own custom evaluation criteria for accurate and reliable evaluations that align with human preferences in a trustworthy manner, we designed and deployed EvaluLLM (Desmond et al., 2024). This tool enables users to generate evaluation outputs by providing a prompt, selecting multiple models, and defining LLM-as-a-Judge with custom metrics using natural language. Users can then review the results and provide feedback, inspecting the agreement between human and AI evaluations through a blind review process. In this paper, we use EvaluLLM as a conceptual design probe with users to explore the design space of how to support development of custom evaluation criteria for accurate and reliable evaluations that align with human preferences in a trustworthy manner.

The overall user flow of EvaluLLM comprises of three stages (see Figure 1). The build experience focuses on defining the LLM-assisted evaluation experience to initiate the auto-evaluation process, the review experience, providing a high-level summary of the evaluation results, and the inspect experience allows users to manually examine the generated outputs through a blind review process. The data generated from this process can be used to calculate the agreement rate, assisting practitioners in better assessing the agreement between human and LLM-as-a-judges. This assessment is crucial for calibrating trust and aids in making informed decisions about whether to change configurations and rerun the evaluation.

In the absence of reference data, related studies suggest that LLMs may not be entirely suitable for use as numerical judges (Zheng et al., 2023). This is because grading based on single answers may fail to detect minor distinctions between specific pairs. Furthermore, the outcomes could become unreliable, as absolute scores tend to vary more than relative pairwise results when there are changes in the judging model (Zheng et al., 2023). To mitigate these challenges, EvaluLLM uses a pairwise comparison approach, as it can reduce the complexity of the evaluation task by breaking down the comparison of multiple outputs into smaller decisions between pairs of data which might yield to more accurate evaluation results at the cost of additional

inference operations. The evaluation method involves making pairwise comparisons between the outputs of different models, similar to the AlpacaEval approach (Li et al., 2023). However, instead of comparing outputs to a single reference, they are compared against one another.

3.1 Build

The build experience (see Figure 1) includes two major components: the Generator (Figure 1A) and the Evaluator (Figure 1B). The Generator section (Figure 1A) is designed to produce evaluation data, supporting users in selecting a pre-uploaded dataset and inputting their task prompts. Users can incorporate data variables from the dataset's structure into the task prompt using the conventional curly bracket format. Additionally, the system provides a range of LLMs for users to choose from for the purpose of performance evaluation. The Evaluator section (Figure 1B) is where users can choose the LLM-as-a-judge model for automatic evaluation and specify the custom metrics that the judge will use to assess the outputs from the generator. This initial version of EvaluLLM, deliberately provides only a freeform input box to support maximum creativity, as the aim was to gain more insights into the types of inputs users would provide to define criteria in natural language and the kind of support users might need to define custom metrics. Once the user completes the setup, they can click the "Run Evaluation" button to initiate the evaluation.

3.2 Review

Upon completion of the automatic evaluation, results are available for review. Users can view a high-level performance summary and a detailed results table. The summary includes a model leaderboard (Figure 1C), ranking selected LLMs by their win rates derived from evaluated output pairs. The performance visualization (Figure 1D) shows detailed win-loss statistics for each model based on pairwise comparisons by the LLM-as-a-judge. Additionally, the agreement rate (Figure 1E) indicates the alignment between human and LLM-as-judges, helping users gauge the reliability of evaluations. This feature becomes available after users manually rate output samples.

3.3 Inspect

Users can examine auto-evaluation results through two main methods. First, users can conduct a blind review, manually inspecting data to assess

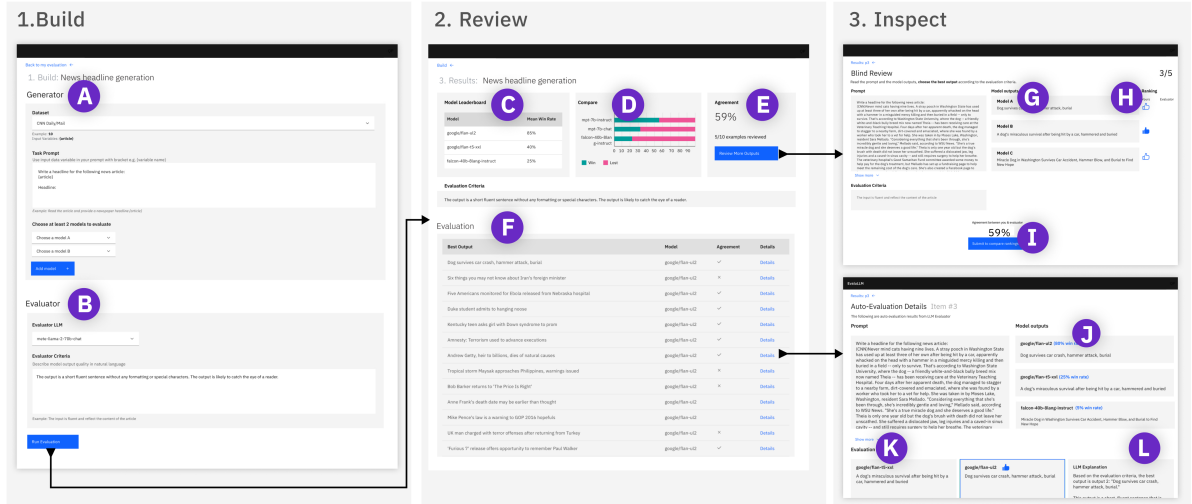


Figure 1: EvaluLLM interfaces and key features

the reliability of LLM evaluations (Figure 1G). In this process, models’ names are hidden to prevent bias, and users select the best output from all presented outputs. Ratings from this process are used to calculate an agreement score, which measures alignment between user and LLM-as-a-judge preferences (Figure 1E, I). After rating, users can view model identities and the updated agreement score (Figure 1H, I), providing insight into the effectiveness of the evaluation criteria. Users can also access detailed results on the review page, which displays the LLM-as-a-Judge’s aggregated rankings and win rates from pairwise comparisons (Figure 1J). Evaluation rationales are provided next to each comparison result (Figure 1L, K), helping users decide whether to trust the results or adjust settings for a reevaluation.

4 Methodology

Our goal was to explore the challenges users encounter during LLM-assisted model evaluations and, based on our observations, to design a framework that meets their needs and supports effective collaboration between humans and LLM-as-a-judges. We used EvaluLLM to facilitate the creation of evaluation tasks and conducted our research through semi-structured interviews using Webex. Participants accessed a prototype of EvaluLLM, shared their screens, and used think-aloud methods to create evaluation tasks. Each participant worked on the same task: using an LLM-as-a-judge to identify the best model for generating headlines from the CNN/Daily Mail dataset.

4.1 Participants

We recruited 8 industry professionals (Appendix Table 1) with deep domain knowledge in model evaluation at a large technology company (2 females and 6 males) via social media recruiting, with participation and recommendations from various individuals. These industry professionals primarily consist of data scientists, software engineers, and AI engineers. Eligible participants were those who had hands-on experience evaluating large language model performance in their projects in the past year. The interviews were conducted remotely, and participants volunteered and consented to the recording of the session, as well as to the use of the interview results for research purposes.

4.2 Data Analysis

Two authors independently reviewed the transcripts from recorded video sessions to pinpoint users’ needs, system shortcomings, and challenges in the evaluation workflow. This independent review helped minimize bias and allowed for a comprehensive data exploration. Each author used a codebook of example quotes to support the identified themes. The authors then met to merge similar themes and address any initially missed, resulting in three main categories: use case challenges, evaluation criteria, and evaluation workflow, detailed in Appendix Table 2. This classification captures the complexities of the evaluation process, encompassing users’ needs, system limitations, and evaluative challenges.

5 Results

Our data analysis identified nine themes, categorized into use case challenges, evaluation criteria, and evaluation workflow (for a full list with example quotes see Table 2 in the Appendix).

5.1 Use Case Challenges

The system requires users to input a prompt for their specific task, after which it generates the output and proceeds with the evaluation. This approach involves sending the identical prompt to various models for output evaluation. However, this methodology poses limitations for experienced users who tailor prompts for specific models, such as LLaMA. Our participants described instances of **absence of specifications** where clients lack clarity on the task’s data requirements.

Additionally, there are numerous open-source and closed-source LLM models available, and users would like to test various setups, e.g., model selections, model configurations, and prompts. They would like the system to **support comparison with different setups**. Given time constraints and limited investment resources, it is often impractical to test all models with their use case data. Teams usually begin **with top-performing models**, either from public benchmarks close to their use case or chosen based on their well-known reputation. Model selection is transient and highly constrained by project requirements. Instead of evaluating multiple models’ performance with different prompts, they typically start with 1-2 models and **improve performance through prompt engineering**. This involves running the model with various prompts and parameter settings, where they often iterate over the setup to match specific baseline performance. It requires rapid performance comparison and support for evaluation data, accommodating multiple models and considering combinations **with different setups**.

Shifting evaluation priority often occurs as the project progresses. At the beginning of the project, where the main purpose is often the proof of concept for a specific proposed solution, the evaluation focus is mainly around feasibility testing. This involves assessing whether the proposed system or solution can produce accurate answers. However, as the project progresses into production, the evaluation purpose might shift from rapid model performance comparison to continual improvement with user feedback, performance monitoring, and report-

ing potential issues to draw developers’ attentions. As evaluation priorities might differ for various use cases in different project phases, when designing an LLM-as-a-Judge solution, shared needs among these different phases and unique requirements in each phase need to be clearly articulated. This could help better define and design the experience and interaction to effectively support the diverse requirements for each phase.

5.2 Evaluation Criteria

We identified several themes related to how users developed, changed, and trusted the evaluation criteria they were working with. While participants appreciated the flexibility of using the freeform approach in EvaluLLM, many expressed that they **desire structured and customizable templates** for specific use cases that can be tweaked for their purposes. They believe such templates would help them start with an evaluation baseline.

Moreover, participants highlighted the necessity of distinct evaluation criteria for various tasks. For example, they noted that a RAG task might require one set of criteria, while a creative task might demand another. Participants often crafted criteria complete with descriptions and scoring. One typical approach involved naming each criterion, defining it, and then assigning a score.

Evaluation criteria serve as a medium to communicate user preferences to the model. An effective criterion not only needs to reflect the user’s preferences but also must function well to enable the model to understand and follow instructions. When reflecting on evaluation criteria, participants expressed the **need for multiple rounds of iterations** when refining their criteria. *"It can be really hard to figure out how to express the evaluation criteria in a way that makes sense to the model. But it can also just be hard in your own mind to figure out what it means for a title to be good."* P2

The importance of giving supporting multiple rounds to refine and expand criteria emerged when looking at the types of dimensions participants created. We found that users tend to prioritize more objective metrics such as accuracy before they start to consider the styling of the outcome. At the beginning of the project, the primary concern for a client is getting the correct answer from the model. That is not to say, that our participants did not care about more subjective criteria, but that happens later in the process.

Although users might have a rough idea of what

they want, it is challenging to describe everything at the beginning, especially when they don't have access to the evaluation data. One participant struggled during the criteria definition process as he was required to define the criteria before he could see the output data. Providing the output might help users articulate what they want or don't want, assisting them in iterating the criteria description or adding examples to better align with their preferences.

Users express a desire for more than just a high-level result summary; they are keen on obtaining a detailed breakdown of each dimension and a need for the system to **display performance for each criteria individually**. EvaluLLM currently only presents a win rate as a high-level performance summary metric to showcase the winning model on the leaderboard. Participants expressed the desire to view performance across each dimension rather than a high level win-rate.

5.3 Evaluation Workflow

While presenting the tool to users, we probed them on their current evaluation workflows and how they would imagine incorporating EvaluLLM. Users expressed the challenges they faced when doing manual evaluations and how they would use automated methods and the EvaluLLM experience to address those challenges. Although there are only 10 examples in our testing dataset, generating the evaluation results after user created the evaluation is time consuming because of calls to the model. Model calls are expensive and time consuming and one potential way to address this is to **run the evaluation on a subset of the data first**.

To evaluate the agreement of the LLM-as-a-Judge preferences with humans, participants were asked to conduct blind reviews of the model's output. These reviews would be utilized to calculate the agreement between the LLM-as-a-judge and the participants. While it is beneficial to observe the agreement rate in the summary page, users also desire more control over the workflow and seek instant feedback during the manual review process. They would like to see how much the LLM-as-a-judge agrees with them once they provide feedback and wish for the system to proactively provide criteria modification suggestions. One way of providing **instant feedback on human-AI agreement** is to allow users to either initially upload human evaluations for comparison with the automatic evaluations. Another way is to conduct a blind review

before the evaluations are presented, ensuring that users receive instant feedback on human-AI agreement as soon as the evaluations are ready.

During testing, we observed that some participants might provide overly detailed instructions for both the task prompt and the evaluation criteria. The design intention was to simplify the user input requirements, seeking only the evaluation criteria rather than a complete evaluation prompt with detailed evaluation process. However, some participants included the step-by-step evaluation process in the criteria definition input. Additionally, some participants inquired about adjusting their evaluations per judge.

As our participants are domain experts in model evaluation, they are well aware of potential biases in the model. They actively seek transparency regarding the bias mitigation strategy to effectively calibrate their trust in LLM-as-a-Judge results. Additionally, participants were cognizant of self-enhancement bias (Zheng et al., 2023) and expressed concerns about the LLM-as-a-judge being one of the models to be evaluated. **Ensuring transparency for trustworthy evaluation** was deemed crucial by users, such as transparency concerning the prompts sent to the judge and whether bias mitigation has been implemented. One user remarked, *"It seems like Granite always displays first, and Flan-UL-2 always comes second. Does the system randomly switch positions?" P5*

5.4 Limitations

Our study is based on a small sample of only 8 domain experts, potentially impacting the generalizability of our findings. In addition, our methodology primarily concentrated on observing users utilizing our specific evaluation tool with one pre-defined dataset. This approach may restrict the broader applicability of our results. Note that EvaluLLM at the time of this study was a functioning proof-of-concept but not yet a scalable systems that can be deployed to a large user population. However, we believe our findings still offer relevant insights into the challenges and needs users encounter when using LLM-as-a-Judge tools, as evidenced by our focused line of questioning aimed at understanding how more automated evaluations integrate into users' workflows.

6 Discussion and Design Recommendations

Our findings highlight user needs across different use cases when using LLM-as-a-judge. Users require guidance to evaluate model outputs effectively. We discuss the implications of our findings and propose design recommendations for LLM-as-a-judge tools and user experiences.

6.1 Efficient Criteria Iteration

LLMs can generate high-quality outputs aligned with human preferences, but processing the entire dataset is costly and time-consuming, especially with methods like pairwise comparisons, which increase compute costs significantly. To optimize efficiency, it's advisable to start a project by allowing users to refine their evaluation criteria using a representative data sample before scaling up to the full dataset (see Figure 2). Effective sampling enhances learning for LLM-as-a-Judge by selecting diverse and representative outputs. Techniques like clustering (Chang et al., 2021) or graph-based search (Su et al., 2022) can aid in output selection for human evaluation. Addressing misalignments and manually reviewing low-confidence outputs (Desmond et al., 2021) are crucial, as is displaying a subset of evaluations to lessen users' cognitive load and facilitate iterative refinement of evaluation criteria.

6.2 Structured and Customizable Templates

For creative generation tasks, it's crucial to employ diverse, custom criteria. To streamline this process, we propose providing standard criteria that are universally applicable across various use cases, supplemented by customizable templates. As illustrated in our design explorations (see Appendix Figure 3), users can select from predefined criteria dimensions (Figure 3A) or utilize recommended templates for common scenarios (Figure 3B). These templates are designed to be flexible, allowing easy adaptation to specific user needs.

Further enhancing customization, the proposed templates support hierarchical organization (see Appendix Figure 4), enabling the addition of new criteria dimensions (Figure 4G), nesting of sub-criteria (Figure 4F), and removal of unwanted elements (Figure 4H). Users can also adjust scoring scales (Figure 4E). This hierarchical structure, supported by findings from related works (Zheng et al., 2023) (Kim et al., 2023c) (Stureborg et al., 2023),

allows users to start with broad criteria and refine them to capture specific task nuances. To foster ongoing improvement and reuse, the system should enable users to save and share these templates (Figure 4B). Considering the benefits of balanced evaluations, users should be able to adjust the weight of different criteria dimensions, aligning more closely with human preferences. The inclusion of reference examples within the templates (Figure 4D) can further refine the criteria based on actual output data, enhancing the preference agreement process. This approach not only makes the criteria definition process more efficient but also ensures consistency and rigor in evaluating creative tasks, leading to more accurate and effective assessments.

Providing structured and customizable templates will not only expedite the process of criteria definition but also foster consistency and rigor in the evaluation of creative generation tasks, which will contribute to more accurate and effective evaluations.

6.3 Interactive Criteria Iteration

Our findings revealed crafting effective criteria typically requires multiple iterations. Criteria components such as name, definition, scale, and examples often need definition and refinement as users evaluate outputs. Users include examples of both poor and excellent outputs to help LLM-as-Judges distinguish quality through few-shot learning techniques. Related work (Kim et al., 2023c) indicates that users often develop new criteria during evaluations. To facilitate this process, a real-time feedback system that allows users to immediately see the impact of criteria modifications would be useful. Additionally, a user-friendly interface that enables easy modification and experimentation with criteria could significantly improve the efficiency and customization of the evaluation process.

6.4 Ensure Consistency

As human preferences may not be consistent within the same set, aligning with frequently changing preferences becomes a challenge. A self-consistency check mechanism can expedite this alignment. When refining criteria, any discrepancies between human and LLM-as-a-Judge evaluations should prompt a review of similar sample data post-calibration. Incorporating an automated consistency checker that flags potential criteria conflicts or inconsistencies could streamline the evaluation process by offering actionable solutions to

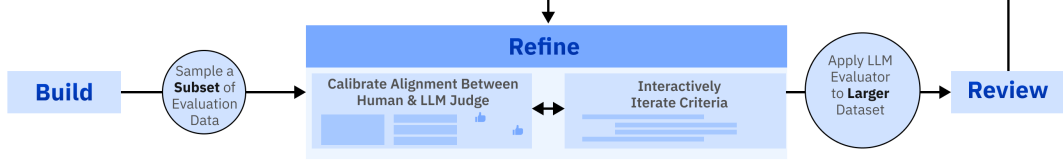


Figure 2: Recommended evaluation workflow: interactive refinement of criteria with a subset of data prior to applying evaluation to entire dataset can potentially improve preference alignment and trust calibration.

address these inconsistencies. Leveraging the diversity of logical paths in complex reasoning tasks, as suggested by recent studies (Stanovich and West, 2000), the self-consistency CoT method (Wang et al., 2023b) can generate multiple reasoning paths, selecting the most consistent answers by averaging over these paths, thus improving evaluation outcomes.

6.5 Support Different Setups

Our findings emphasize the need for an LLM to function flexibly as a judging system throughout different project phases. It should support a variety of evaluation data configurations, including diverse model selections, prompts, and settings. While some evaluations may only compare outputs from a specific prompt and model setting, optimal performance often requires tailored prompts and settings for each model, involving substantial prompt engineering and comparison of different configurations. Thus, the system must not only evaluate common settings across various models but also assess various prompts and settings for select models, highlighting the importance of designing an adaptable LLM judging system.

6.6 Adaptable Reference-Based Evaluation

Our user study findings showed that users often start projects without clear objectives, resulting in evaluations lacking reference data. Users interacting with the LLM-as-a-Judge system gradually accumulate reference data, either directly or from external sources, so it could be beneficial to design systems that incorporate human input to refine preference correspondence using expert-labeled data (Liu et al., 2023b) or other collected references. This flexible approach enhances the system’s effectiveness and trustworthiness, ensuring it evolves in line with user preferences.

6.7 Enhance System Transparency

Our findings indicate that users value transparency to comprehend the LLM’s role as a judge. This

encompasses access to essential details like the specific prompt used (illustrated in Figure 5A) and the implementation of bias mitigation strategies. To design an effective LLM-as-a-Judge system, it is critical to make such information readily available. This can be facilitated by allowing users to view the prompt, enabling the system to explain evaluation results, and integrating visualization tools that demonstrate how user inputs affect the evaluation process.

6.8 Proactively Mitigate Potential Bias

Considering the persistent challenge of bias, systems should implement bias mitigation strategies that include swapping answer order to reduce position bias (Zheng et al., 2023) and treating inconsistent results as ties, or by randomly assigning positions in large datasets (Li et al., 2023) (Zheng et al., 2023). For verbosity bias, the "repetitive list" attack technique (Zheng et al., 2023) challenges LLMs to favor clarity over length in responses. Furthermore, enhancing LLMs’ abilities in mathematical and reasoning tasks can be achieved through Chain-of-Thought approaches (Wei et al., 2022), coupled with reference-guided evaluation where the LLM generates and then evaluates its own initial responses.

6.9 Explore Further Automation

Our study found that task prompts often contain criteria, suggesting the possibility of extracting them automatically for tailored guidelines. Related work also shows that users prefer automated prompt refinement over manual revisions (Kim et al., 2023c). Various suggestions (see Appendix Figure 5), such as rephrasing (Figure 5A), adding reference examples (Figure 5B), incorporating more scales (Figure 5C), and introducing additional dimensions (Figure 5D), could be proactively provided by the system for humans to review to further accelerate evaluation correspondence. While these areas show promise for further improving the efficiency of preference correspondence, considering the lim-

itations of automation systems, it is essential to place humans in the loop to calibrate accuracy and trustworthiness.

7 Conclusion

We studied EvaluLLM, an AI-assisted tool utilizing LLMs alongside humans as judges for LLM-generated content. Our findings highlight the potential of LLMs as customizable judges and underscore the importance of interactive, transparent, and user-centered evaluation processes. Based on our findings, we offer design suggestions for practitioners that can help them build more effective, nuanced, adaptable, and user-friendly evaluation tools that meet diverse needs as compared to automated benchmarks. Inspired by our user research, we are currently in the process of rolling out an evolved AI-assisted evaluation tool to a larger user population to observe "usage in the wild."

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A Participant Information

Table 1 shows the details of participants involved in the user study, predominantly comprising of industry experts such as data scientists, software engineers, and AI engineers. These professionals have practical experience in evaluating the performance of large language models in their projects over the last year.

B Summary of Evaluation Themes and Examples

Table 2 provides further details on evaluation themes generated from the user study, along with corresponding examples from participants’ quotes.

C Recommended Designs

Figure (3)(4)(5) show design examples to help illustrate corresponding design recommendations.

D EvaluLLM Evaluation Workflow

Figure (6) shows the high-level overview of the EvaluLLM workflow, which consists of a Build, Review, and Inspect process.

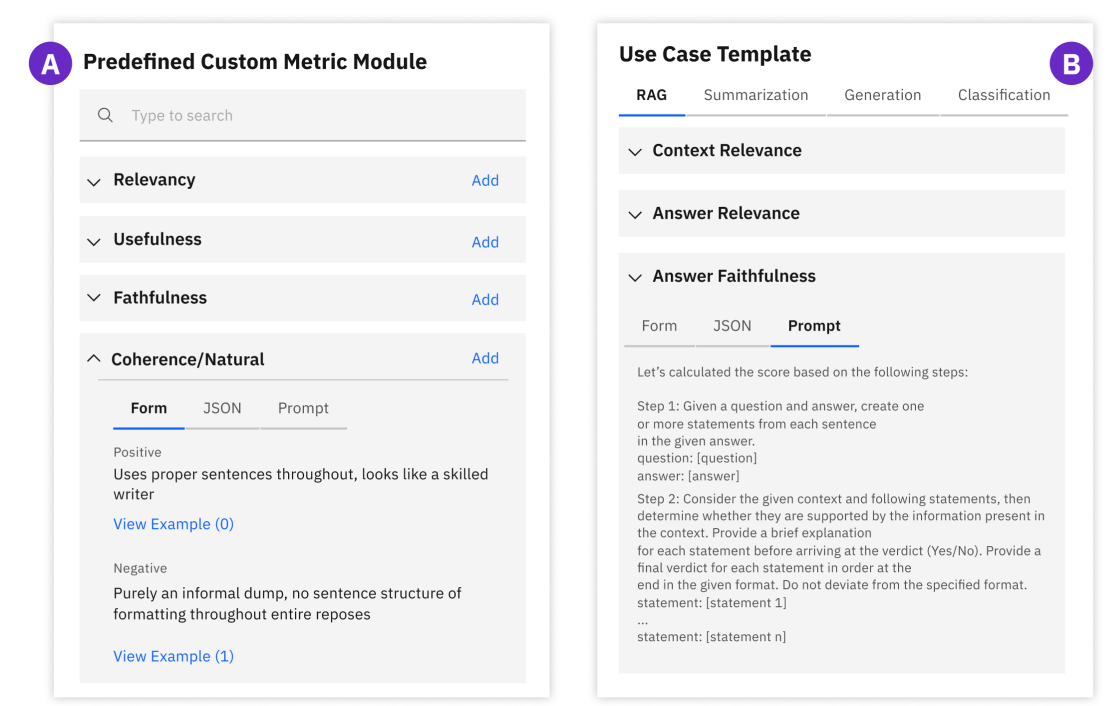


Figure 3: Recommended design to (A) enable users to choose from a list of predefined custom metric modules and (B) enable users to create a set of evaluation criteria based on common use cases.

<i>ID</i>	<i>Gender</i>	<i>Job Role</i>
P1	Male	Lead Software Engineer/Data Scientist
P2	Male	Principle Data Scientist
P3	Male	Lead Software Engineer/Data Scientist
P4	Male	Data Scientist
P5	Male	AI Engineer/Data Scientist
P6	Female	Data Scientist
P7	Male	Senior Technical Manager/Data Scientist
P8	Female	Data Scientist

Table 1: Demographic information from participants in our user study.

Table 2: Table of evaluation themes and corresponding examples. Themes are grouped into three categories: use case challenges, evaluation criteria, and evaluation workflow. Quotes are provided to delineate themes.

Group	Theme	Example
Use Case Challenges	Absence of Specifications	<i>“So we can compare using, metrics such as or BLEU, And this is like this other scenario, which unfortunately is more common, which is client doesn’t even know what they want.” - P5</i> <i>“It was like eighty-twenty, eighty percent of the time they don’t have it.” - P5</i>
	Support Comparison with Different Setup	<i>“Say we had five different models and for each model we had 20 different configurations or something like that. Now that’s 100 different combinations. Um, we’d like the limited judge to be to run on like all hundred. Give us an overview. Which are the three that are actually worth looking at?” - P2</i> <i>“GPT 4 as a baseline and we’re just trying to see how close are we getting with these other models in order to replicate the performance.” - P7</i>
	Shifting Evaluation Priority	<i>“I know that’s like a terrible metric [confusion matrix] to be used as the first one, but we have actually done this with a client because they asked us to do so. They’re looking for just accuracy.” - P5</i> <i>“GPT 4 as a baseline and we’re just trying to see how close are we getting with these other models in order to replicate the performance.” - P7</i>
	Desire Structured and Customizable Templates	<i>“A freeform text box is too simple. I would love there to be templates that I can utilize. And at the very least, be able to just edit so that I can get into my use case.” - P7</i> <i>“More examples might be nice.” - P2</i>
	Need for Multiple Rounds of Iterations	<i>“It can be really hard to figure out how to express the evaluation criteria in a way that makes sense to the model. But it can also just be hard in your own mind to figure out what it means for a title to be good.” - P2</i> <i>“If I think, without having a clearer sense of what the evaluation is, sort of what a baseline evaluation is, it might be nice to have a couple of features of an evaluation that we could just select in like a checkbox.” - P3</i>
	Display Performance for each Criteria Individually	<i>“There might be times where you have to trade off on certain kinds of things and Win rate is not necessarily the best metric because there are multiple categories to define what it means to win.” - P7</i> <i>“So I’m covering a lot of ground there, and I know that’s hard for the model to deal with because now the model has to have a whole lot of different criteria, and it’s all drawn up by the ones, but that’s kind of what a good title headline is about.” - P7</i>
Evaluation Workflow	Run Evaluation on Subset of Data First	<i>“We don’t have a problem here because the data set is small. But, like, if there’s like, a 1000. Then it would it make sense to go through the entire batch and we find out your volume criteria needs to be tweaked.” - P2</i> <i>“I’d want to iterate on my judge enough for it to get a decent annotator agreement and then let it go wild.” - P2</i>
	Instant Feedback on Human-AI agreement	<i>“Tell me when to quit.” -P1</i>
	Ensuring Transparency for Trustworthy Evaluation	<i>“So I definitely want, as we discussed earlier, a lot of transparency and exactly what is being sent to the models to generate the responses and then what is then being sent to the LLM as a judge.” - P2</i> <i>“Maybe a small note on, like, you know what the prompt is, like, what the data set is and what the tool is doing.” - P8</i>

Hierarchical Customizable Template

A

Form

JSON

Prompt

Evaluation Use Case Name

B

News headline generation evaluation

C Weight 20%	Metric Name fluency Positive The output text is well-written, clear, and concise. D View Example (0) Negative The output text is poorly written, confusing, or wordy D View Example (0) E Add more scale +	H
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Weight	Metric Name	
50%	well-written	⌵
Weight	Metric Name	
20%	clarity	⌵
Weight	Metric Name	
30%	concise	⌵

F

Add sub-criteria +

Weight	Metric Name	
80%	Accuracy	⌵

G

Add new criteria +

Figure 4: Recommended design to provide structured and customizable templates that support hierarchical, multi-dimensional evaluations.

Evaluation Criteria

Self Critique

Weight 20%	Metric Name fluency	⌵
Positive The output text is well-written, clear, and concise View Example (0)		
Negative The output text is poorly written, confusing, or wordy View Example (0)		
Add more scale +		

Weight 50%	Metric Name well-written	⌵
Weight 20%	Metric Name clarity	⌵
Weight 30%	Metric Name concise	⌵

Add sub-criteria +

Weight 80%	Metric Name Accuracy	⌵
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Add new criteria +

Iteration Suggestions

Rephrasing:

the seamless and natural flow of language that is not only grammatically correct and coherent but also stylistically polished

Add example

Adding this output as a negative example might improve alignment

Add more scoring scale

Consider using a 1-5 scale to capture more nuanced differences

Add more dimension

Relevancy

Usefulness

Figure 5: Recommended design demonstrating the ability of users to leverage LLM-as-a-Judge for Criteria Iteration.

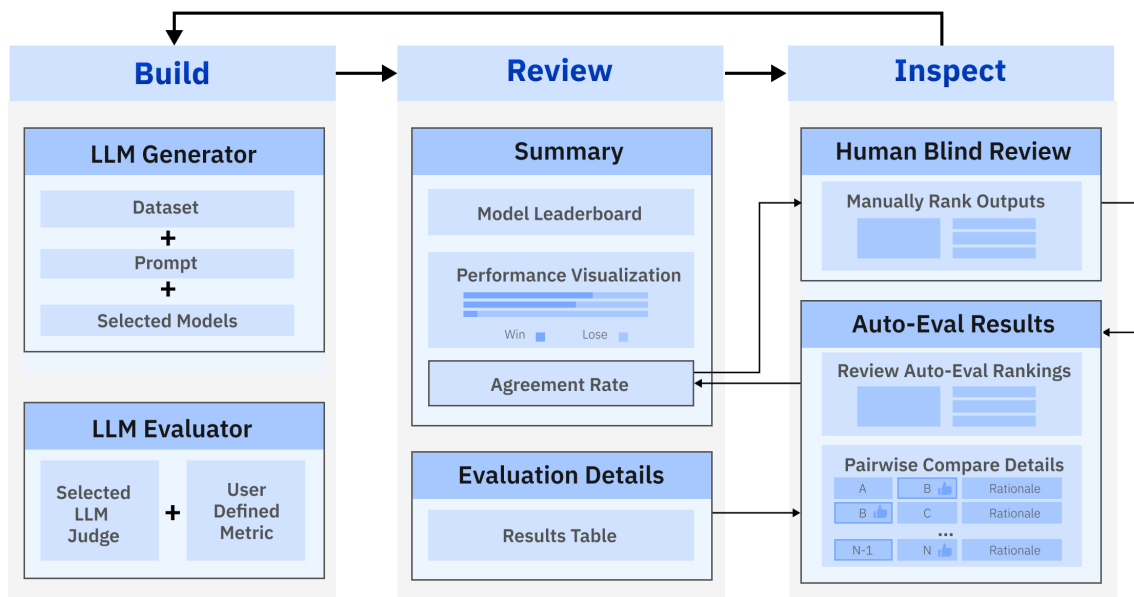


Figure 6: EvaluLLM evaluation workflow overview which consists of a Build, Review, and Inspect process.